

Prueba Objetiva 3

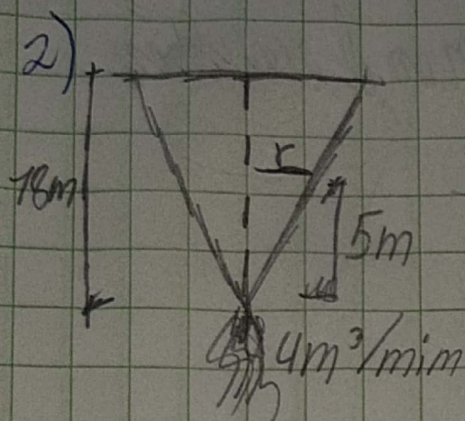
$$1) f(x) = \frac{(e^x + 5)^4 (x^2 + 3)^5}{(e^x - 5)^3 (x + 2)^3}$$

$$\ln y = \ln(e^x + 5)^4 (x^2 + 3)^5 - \ln(e^x - 5)^3 (x + 2)^3$$

$$\ln y = \ln(e^x + 5)^4 + \ln(x^2 + 3)^5 - \ln(e^x - 5)^3 - \ln(x + 2)^3$$

$$\frac{y'}{y} = \frac{4(e^x + 5)^3 e^x}{(e^x + 5)^4} + \frac{5(x^2 + 3)^4 2x}{(x^2 + 3)^5} - \frac{3(e^x - 5)^2 e^x}{(e^x - 5)^3} - \frac{3(x + 2)^2}{(x + 2)^3}$$

$$\frac{y'}{y} = \frac{4e^x}{e^x + 5} + \frac{10x}{x^2 + 3} - \frac{3e^x}{e^x - 5} - \frac{3}{x + 2}$$



$$\frac{r}{5} = \frac{7.5}{18} =$$

$$r = 2.0833 \text{ m}$$

$$Q_{\text{cono}} = Q_{\text{entrada}} - Q_{\text{salida}}$$

$$\pi r^2 \frac{dh}{dt} = Q - 4 \text{ m}^3/\text{min}$$

$$\pi (2.0833)^2 (5 \text{ m/min}) = Q - 4$$

$$Q = 72.78 \text{ m}^3/\text{min}$$

$$3) f(x) = \sin^2 x + \sin x$$

$$D = (0, 2\pi)$$

$$f'(x) = 2 \sin x \cos x + \cos x = 0$$

$$\cos x (2 \sin x + 1) = 0$$

$$\cos x = 0$$

$$x = \cos^{-1} 0 = \pi/2$$

$$2 \sin x + 1 = 0$$

$$x = \sin^{-1}(-1/2) = \pm \frac{\pi}{6} + n\pi$$

$$x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

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x	$f'(x)$	$f(x)$	
0, $\pi/2$	+	2	→ Max.
$\pi/2, 7\pi/6$	-	$-7/4$	→ Min.
$7\pi/6, 3\pi/2$	+	0	
$3\pi/2, 11\pi/6$	-	$-7/4$	→ Min.
$11\pi/6, 2\pi$	+		

$$4) V = \frac{4}{3} \pi r^3 = \frac{4\pi}{3} (1\text{cm})^3 = \frac{4\pi}{3} \text{cm}^3$$

$$\frac{dv}{dr} = 4\pi r^2 \frac{dr}{dr}$$

$$\frac{dv}{dr} = 4\pi (1\text{cm})^2 (2\text{cm}) = 8\pi \text{cm}^3 \rightarrow \text{Error on } d \text{ volumen}$$

$$\frac{dv}{V} = \frac{8\pi}{\frac{4}{3}\pi} = 6 \rightarrow \text{error relative}$$